

Tutorial and example datasets for the INSPECTA software package

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Introduction

Obtaining SDHDF example files

The following table provides example data sets from the Parkes telescope in the SDHDF format. The files can be obtained using the filename download option available from <https://atoa.atnf.csiro.au/listDownload.jsp>.

Summary	File	Description and notes
Bright OH maser	uwl_210722_012146.hdf	Relatively small file (588MB) with 262144 channels/sb for all 26 sub-bands. Contains strong OH maser. Noise source switching
Scan across bright continuum source	uwl_210722_200112.hdf	Scan across 0407-658
On-off pointing of a bright unpolarised continuum source	uwl_210722_212808.hdf uwl_210722_213343.hdf	ON and OFF source observations of the calibrator PKS 0407-658
On-off pointing of a bright, polarised continuum source	uwl_210722_202637.hdf uwl_210722_203712.hdf	ON and OFF observations of 2323-407, a continuum source with known polarisation properties.
Observation of the S8 IAU HI calibration region	uwl_210722_205546.hdf	Observation of the S8 IAU calibration region

Observation of HI galaxy	uwf_210722_191847.hdf uwf_210722_192923.hdf	Observations on and off source for the HI galaxy: NGC7361.
Very early data file		
Non-UWL SDHDF file		
Azimuthal scan		
Data containing band saturation		
Modern, relatively clean data set	2021-08-25-23:23:49	Observation obtained when the site was shut-down for COVID reasons. No visitors and very few people on site.
Data file in the repo		

Obtain the software from https://bitbucket.csiro.au/scm/cpda/sdhdf_tools.git and use the

In all cases the \$SDHDF_RUNTIME environment variable needs to be set and pointing to the 'runtime'.

An easy way to inspect an SDHDF file is to use "silx view"

```
silx view <filename.hdf>
```

Use case: On-off observations

We make use of observations of a region in the Orion nebula carried out as P595 by Macquarie University students (supervised by Jo Dawson).

We can first identify the ON and OFF source pointings using `sdhdf_describe`:

```
$ sdhdf_describe *.hdf
```

```
-----
File          Beam UTC          SDHDF PID  Sched_ID Source      Tel      Observer  RCVR
Bands POL_TYPE CAL  M_Time RA_s    DEC_s
-----
uwl_200915_194251.hdf      0  2020-09-15-19:42:51  1.9.2 P595  16673  Pink_Maser_ON      Parkes  cap056
UWL  26  AABBCRCI  ON  597.7 05:03:21.20 +02:03:04.60
uwl_200915_200436.hdf      0  2020-09-15-20:04:36  1.9.2 P595  16675  OFF_MAIN           Parkes  cap056
UWL  26  AABBCRCI  ON  597.7 05:35:05.73 -02:13:58.20
```

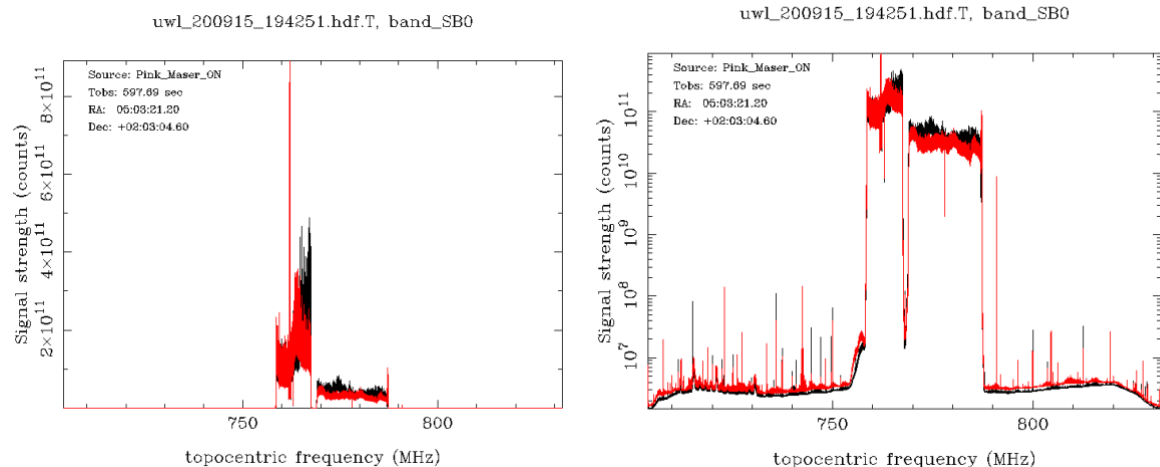
The first file is “on” source. The second file is “off” source. The files are large (>6GB in volume) and so let’s start by average the data in time:

```
$ sdhdf_modify -T -e T *.hdf
```

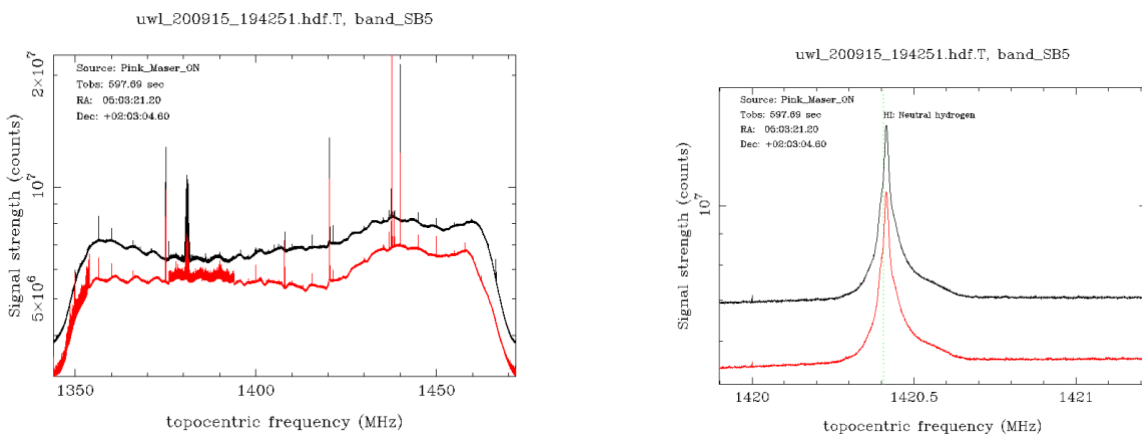
This produces `uwl_200915_194251.hdf.T` and `uwl_200915_212041.hdf.T`. We can now plot the “on” spectrum:

```
$ sdhdf_plotSpectrum -f uwl_200915_194251.hdf.T
```

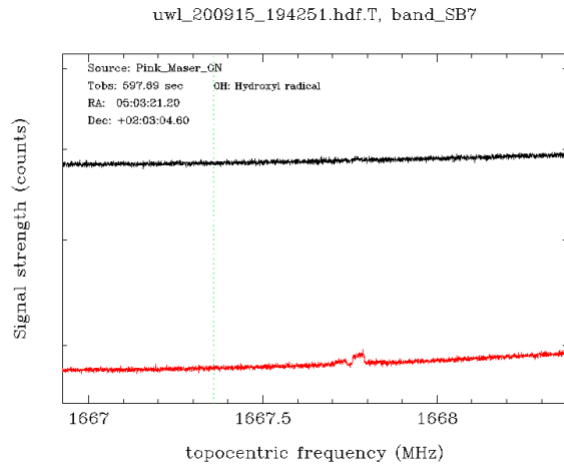
The output spectrum is as follows (left panel). Note that we’re viewing the first sub-band (which has a lot of radio interference) on a linear scale. If we press ‘l’ then we can switch to a logarithmic y-scale (see right panel). The black and red traces represent AA and BB polarisation channels respectively.



This sub-band is hard to work in, so let's use the '>' symbol to move through the sub-bands until we reach sub-band 5 (left panel below). We can zoom in around the HI line by moving the mouse cursor slightly to the left of 1420 MHz, pressing 'z', then moving the mouse to the right and clicking the mouse cursor (right panel). If we press 'm' we can plot the position of the rest frequency of the spectral line (as a vertical, green dashed line).



If we zoom-in around the OH line at 1667 MHz we see a signal that is of interest:

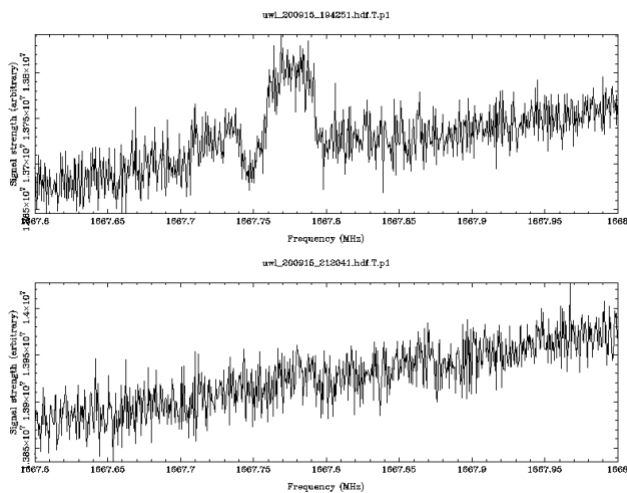


We can sum the polarisations using

```
$ sdhdf_modify -p1 -e p1 *.T
```

And then determine whether the signal is, or is not, in the off source pointing:

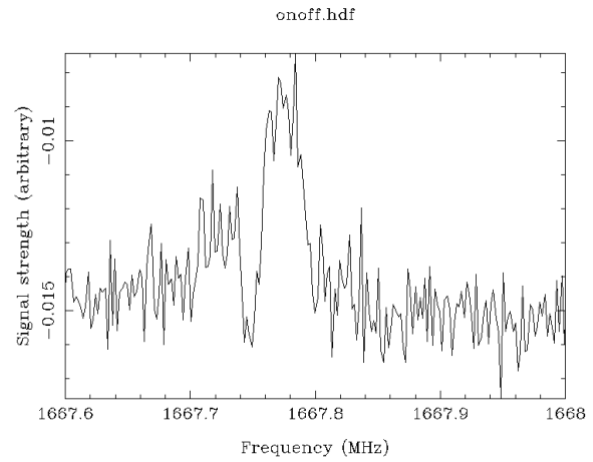
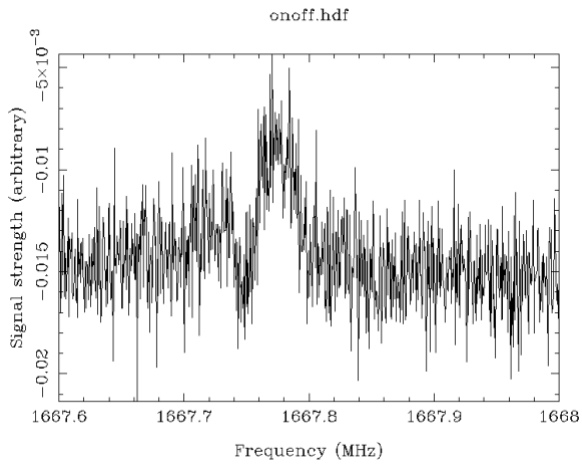
```
$ sdhdf_plotMultiSpec -sb 7 -f0 1667.6 -f1 1668 -ny 2 uwl_200915_194251.hdf.T.p1 uwl_200915_212041.hdf.T.p1
```



This seems convincing. We can form the quotient $[(\text{ON}-\text{OFF})/\text{OFF}]$ using:

```
sdhdf_onoff -on uwl_200915_194251.hdf.T.p1 -off uwl_200915_212041.hdf.T.p1 -o onoff.hdf
sdhdf_plotMultiSpec -sb 7 -f0 1667.6 -f1 1668 onoff.hdf
```

The resulting profile is shown in the left-hand plot below. We can include the command line argument `-av 4` when using `sdhdf_plotMultiSpec` to average 4 adjacent frequency channels together (right-hand panel):



Note that in this simple example we have not carried out many standard processing tasks including calibration, converting to the local standard of rest (and velocity), RFI flagging etc.

Use case: Frequency averaging after RFI removal

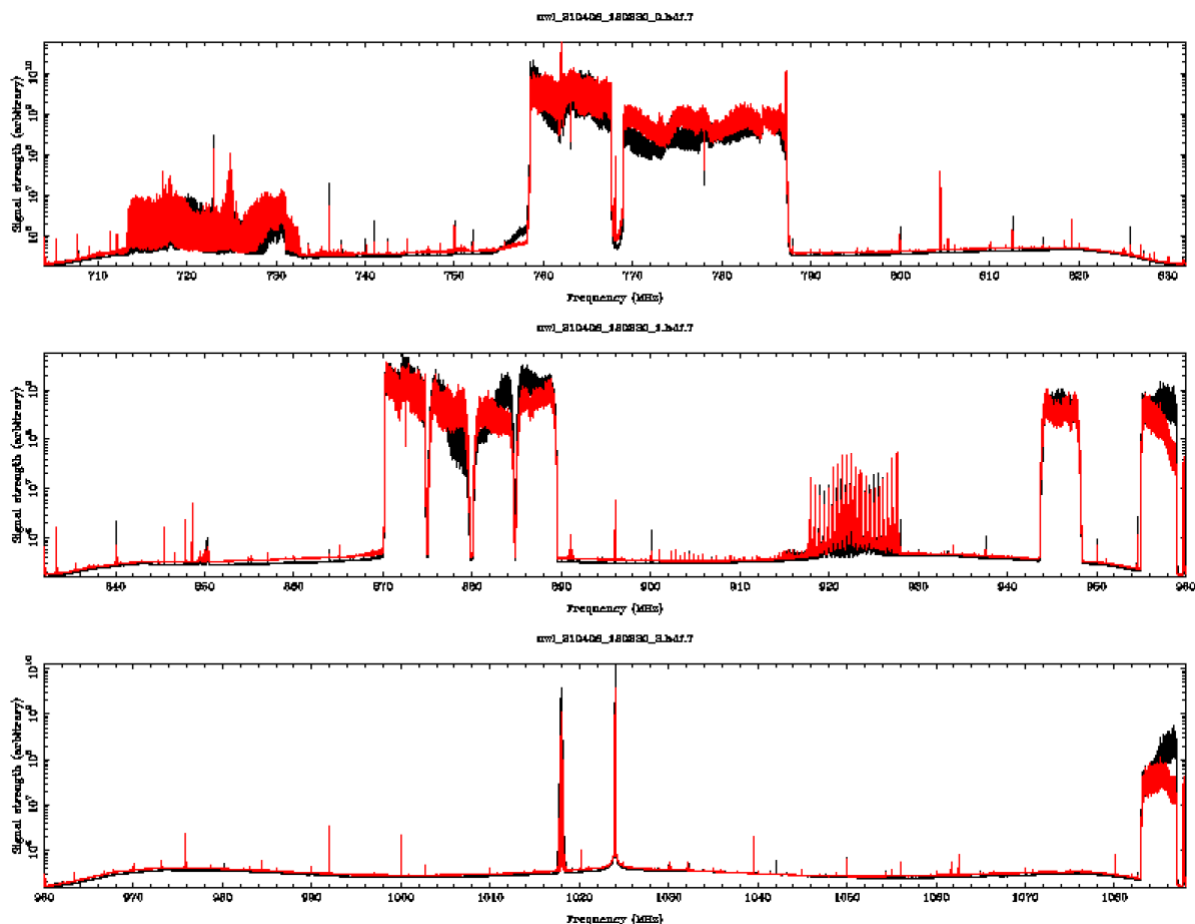
The aim here is to remove RFI automatically from a given data set, then average in frequency to 1 MHz channels and then determine whether further, manual, RFI flagging is required.

Three example data files were obtained from the P1117 observing project. The observations are large and so these files have been (automatically) split by frequency band. The three files correspond to the lowest three frequency bands.

To identify what the files look like without any RFI flagging or frequency averaging we run:

```
$ sdhdf_modify -T -e T uwl*.hdf [to average the data files in time]
$ sdhdf_plotMultiSpec -log -ny 3 uwl_*.T
```

This makes the following plot:



These observations were obtained from Murriyang and so the relevant flagging information is stored in:

```
$SDHDF_RUNTIME/observatory/parkes/rfi
```

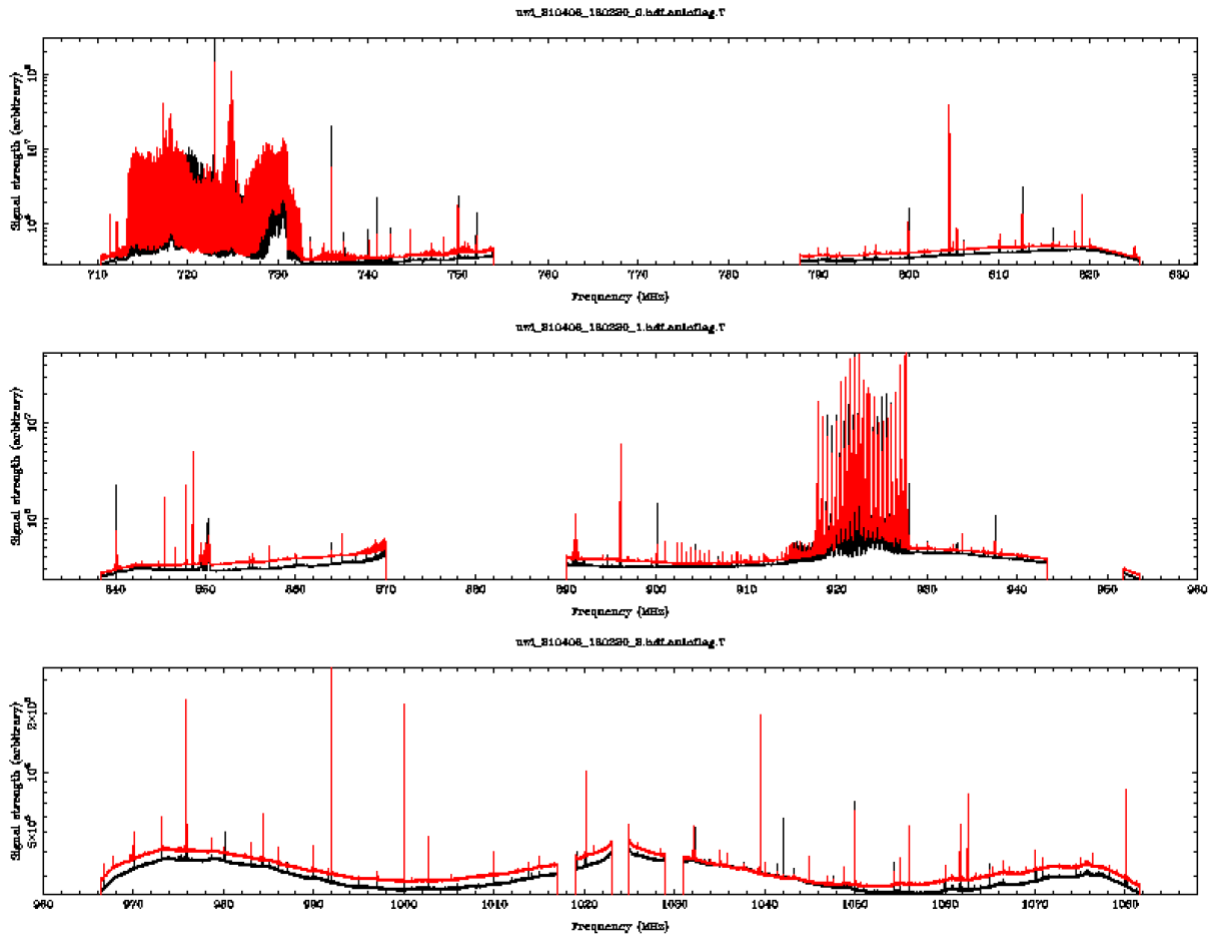
That directory contains “persistentRFI.dat” which provides the frequency bands corresponding to persistent RFI at the Parkes observatory.

The `sdhdf_autoFlag` code is used for automatic flagging.

```
$ sdhdf_autoFlag -edge 5 uwl_*.hdf
```

The `-edge 5` command line option indicates to remove 5% of the sub-band edges. This is used for the current Parkes UWL as the band edges contain aliasing effects. This command produces a set of files with extension `autoflag` in the local directory. We can then time average these files and plot to compare with those above:

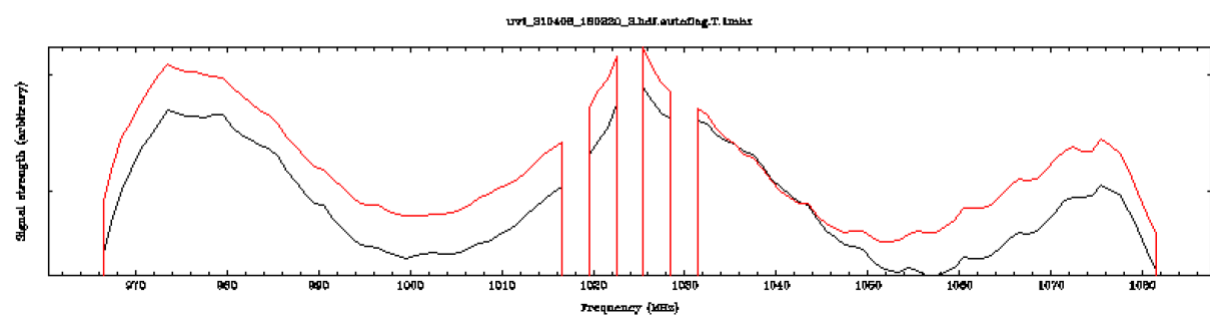
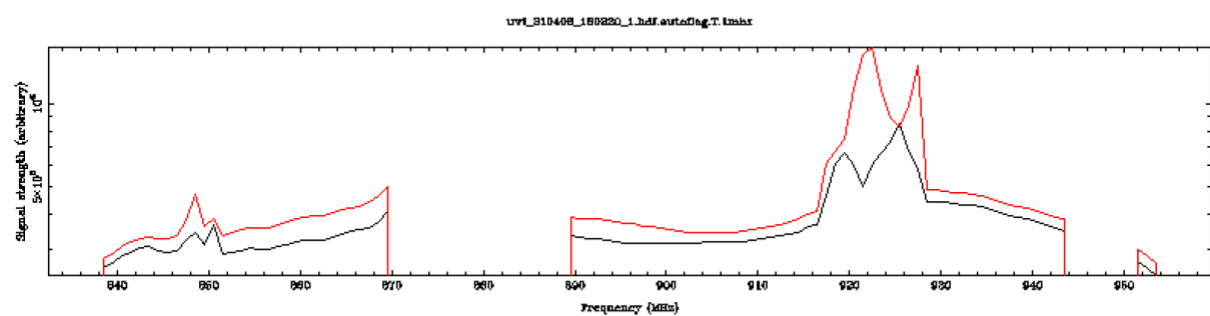
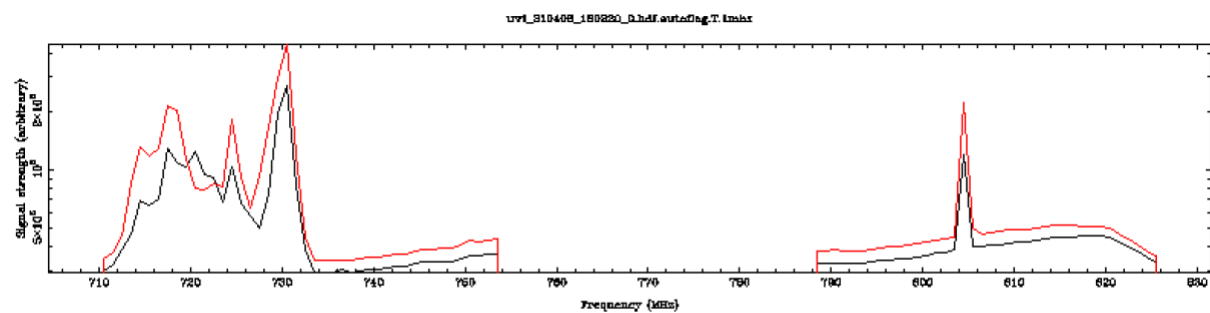
```
$ sdhdf_modify -T -e T uwl*.autoflag  
$ sdhdf_plotMultiSpec -log -ny 3 uwl_*.autoflag.T
```



We can see that the strong RFI signals (primarily from mobile transmission towers) have been removed. This is clearly not perfect, but let us now average in frequency to form 1 MHz channels:

```
$ sdhdf_modify -fav 2048 -e 1mhz uwl_*.autoflag.T
$ sdhdf_plotMultiSpec -log -ny 3 uwl_*.autoflag.T.1mhz
```

This gives:



Use case: Having a play with a scan across a source

Request: The user wishes to explore the SDHDF format and software using a scan across a bright continuum source. We will use a P737 scan across PKS 0407-658.

Note: This scan was a great circle scan. Dhagu allows different types of scans.

Let us first check the basic properties of this file:

```
$ sdhdf_describe uwl_200720_024617.hdf
```

This gives basic information:

File				Beam	UTC	SDHDF		PID	Sched_ID	Source	Tel
Observer	RCVR	Bands	POL_TYPE	CAL	M_Time	RA_s	DEC_s				

uwl_200720_024617.hdf				0	2020-07-20-02:46:17	1.9.1	P737		14710	0407scan_ROW5	Parkes
hob044	UWL	26	AABBCRCI	ON	116.9	04:03:50.81	-65:40:11.50				

We can see that the source name is 0407scan_ROW5, the project code (P737), that 4 polarisation products were recorded (AABBCRCI) and that the calibrator was ON. The integration time was 116.9 seconds.

To find out the sub-bands stored in the file we use the -band option:

```
$ sdhdf_describe -band uwl_200720_024617.hdf
```

This gives:

#	BandID	F0	F1	NCHAN	TDUMP	NPOL	NDUMP	TOBS			

[Band]	000 band_SB0	704.00	832.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	001 band_SB1	832.00	960.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	002 band_SB2	960.00	1088.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	003 band_SB3	1088.00	1216.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	004 band_SB4	1216.00	1344.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	005 band_SB5	1344.00	1472.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	006 band_SB6	1472.00	1600.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	007 band_SB7	1600.00	1728.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	008 band_SB8	1728.00	1856.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	009 band_SB9	1856.00	1984.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	010 band_SB10	1984.00	2112.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	011 band_SB11	2112.00	2240.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	012 band_SB12	2240.00	2368.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	013 band_SB13	2368.00	2496.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	014 band_SB14	2496.00	2624.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	015 band_SB15	2624.00	2752.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	016 band_SB16	2752.00	2880.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	017 band_SB17	2880.00	3008.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	018 band_SB18	3008.00	3136.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	019 band_SB19	3136.00	3264.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	020 band_SB20	3264.00	3392.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	021 band_SB21	3392.00	3520.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	022 band_SB22	3520.00	3648.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		
[Band]	023 band_SB23	3648.00	3776.00	32768	0.999	4	117	116.933	uwl_200720_024617.hdf		

```
[Band] 024 band_SB24      3776.00  3904.00  32768    0.999    4    117   116.933  uwl_200720_024617.hdf
[Band] 025 band_SB25      3904.00  4032.00  32768    0.999    4    117   116.933  uwl_200720_024617.hdf
```

Showing that 26 sub-bands were recorded with the spectral dump time of 1 second.

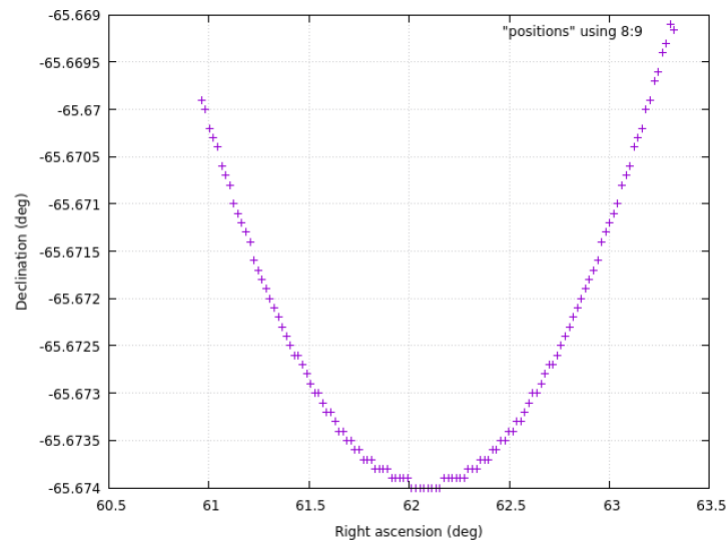
In particular for a scan we would like to know exactly where we were pointing at the centre of each spectral dump. We can obtain this with the `-dump` command line argument:

```
$ sdhdf_describe -dump uwl_200720_024617.hdf
```

```
-----
      ElapsedTime MJD          UTC          AEST          RA          DEC          RADEG  DECDEG  AZ          EL          GL          GB
-----
[SDUMP]      0.500 59050.1154800 02:46:17.50 12:46:18.40 04:03:50.81 -65:40:11.5  60.962 -65.670 209.390  39.053 278.829 -41.327
[SDUMP]      1.499 59050.1154920 02:46:18.50 12:46:19.40 04:03:55.66 -65:40:12.0  60.982 -65.670 209.390  39.059 278.825 -41.319
[SDUMP]      2.499 59050.1155030 02:46:19.50 12:46:20.40 04:04:00.51 -65:40:12.5  61.002 -65.670 209.390  39.067 278.820 -41.312
[SDUMP]      3.498 59050.1155150 02:46:20.50 12:46:21.39 04:04:05.35 -65:40:13.0  61.022 -65.670 209.390  39.075 278.815 -41.304
[SDUMP]      4.497 59050.1155270 02:46:21.50 12:46:22.39 04:04:10.20 -65:40:13.5  61.042 -65.670 209.390  39.080 278.810 -41.297
[SDUMP]      5.497 59050.1155380 02:46:22.50 12:46:23.39 04:04:15.05 -65:40:14.0  61.063 -65.671 209.390  39.087 278.805 -41.289
...
```

We can store the positions in a text file using:

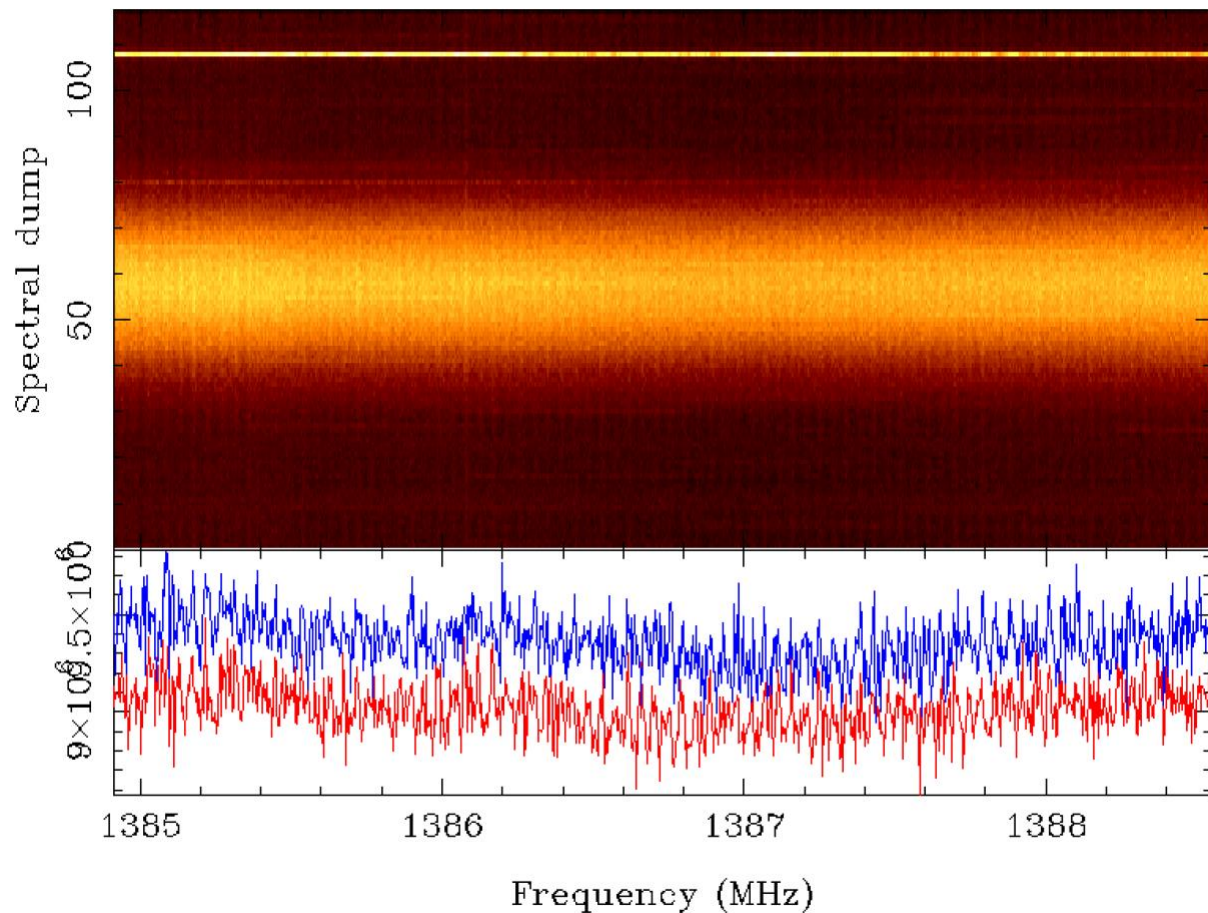
```
$ sdhdf_describe -dump uwl_200720_024617.hdf | grep SDUMP > positions
```



We can make a simple waterfall plot using `sdhdf_waterfall`:

```
$ sdhdf_waterfall -band 5 uwl_200720_024617.hdf
```

Here we selected a relatively clean band (band 5) and then used 'x' to zoom-in on the x-axis to a clean part of that sub-band. The result is:



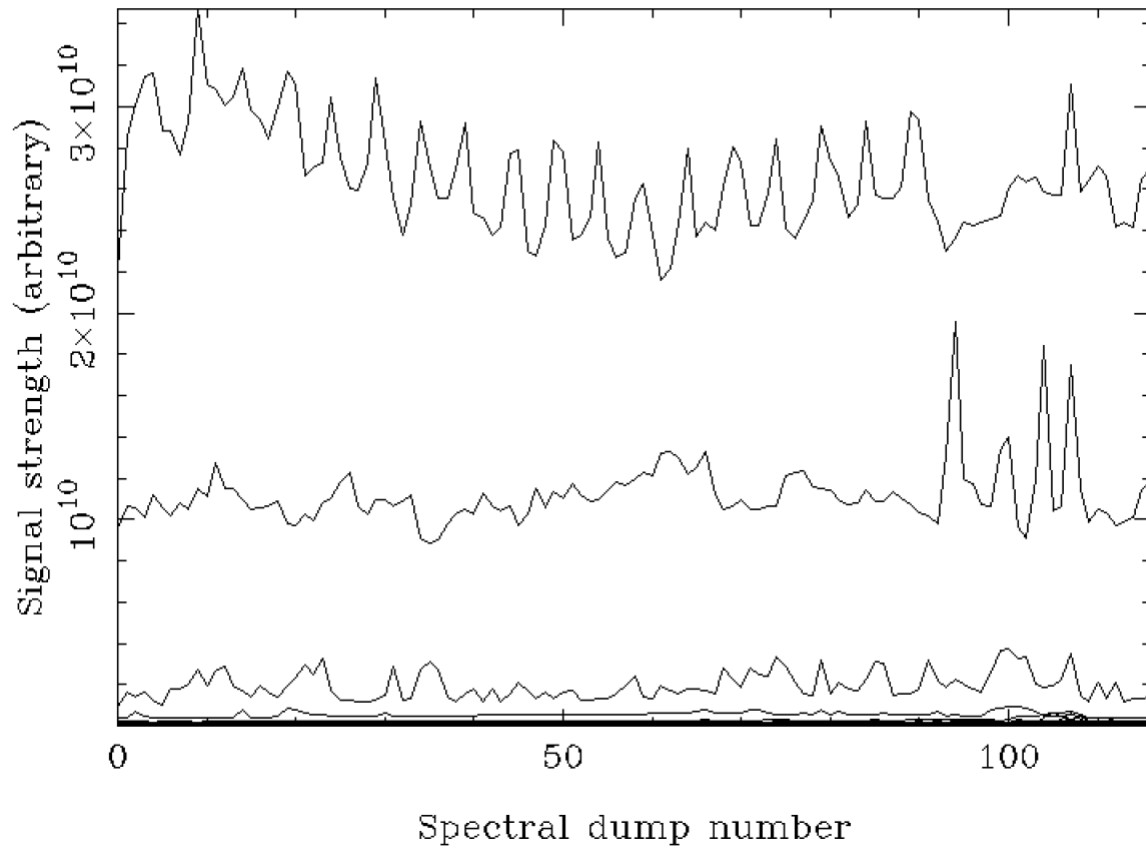
The top panel shows the uncalibrated signal strength as a function of frequency and time (colour scale). The bottom panel shows the min-hold and averaged spectrum averaged over these spectral dumps.

For a more detailed look at the scan we can use `sdhdf_plotScan`:

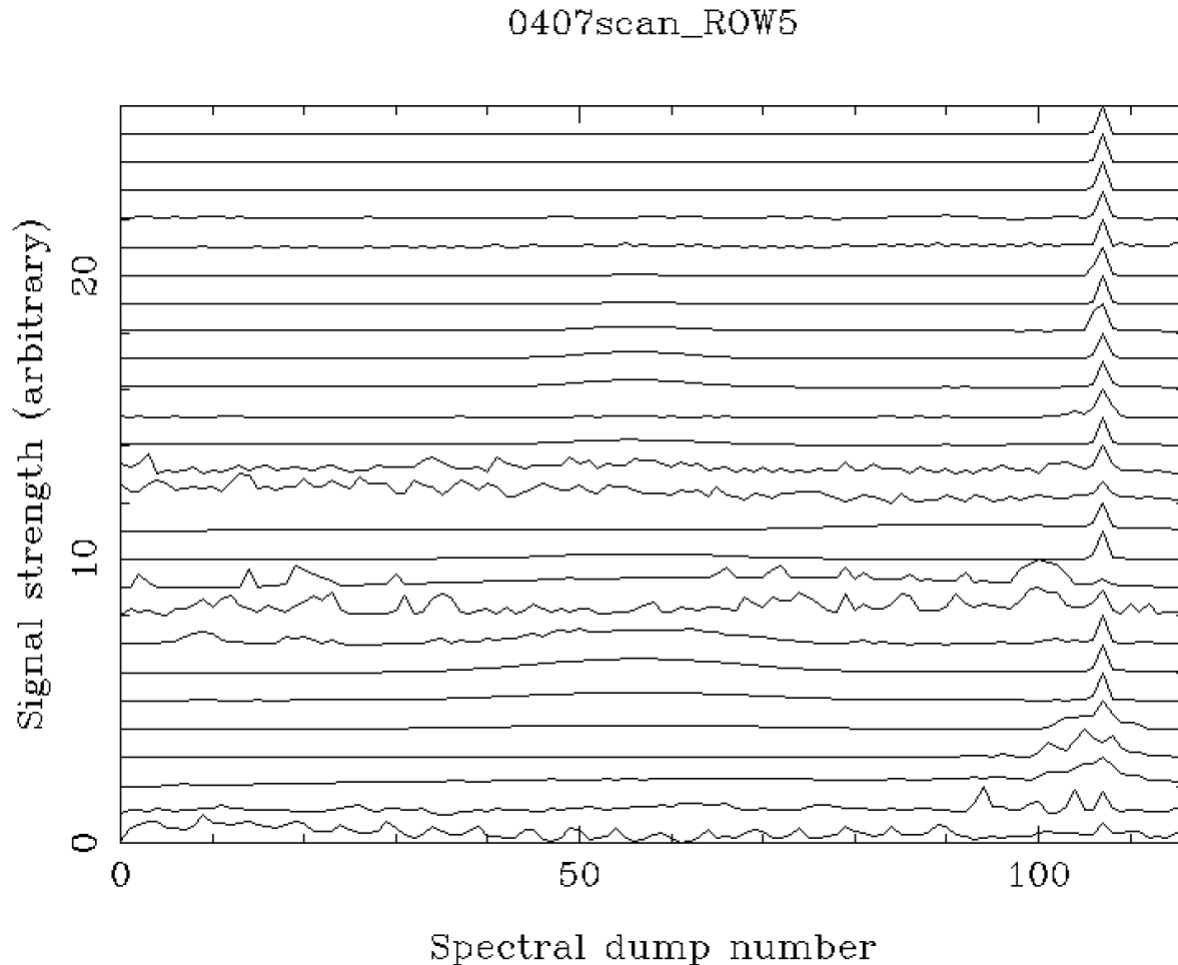
```
$ sdhdf_plotScan -f uwl_200720_024617.hdf
```

The output is shown below. The x-axis is the spectral dump number and the y-axis is the signal strength. The data in each sub-band is averaged in frequency and then plotted. Clearly some sub-bands are at a much higher level than other bands.

0407scan_ROW5



Pressing 'n' allows us to normalise each plot and then pressing 'o' offsets each band from each other:



The “triangular” feature on the right hand side is broad-band RFI that lasted for one spectral dump. The slight “mound” in the middle is the continuum source. Pressing ‘z’ allows you to use the mouse to zoom in on a region and press ‘u’ to unzoom back to the original plot. Pressing ‘x’ toggles the x-axis from spectral dump number to various other parameters (e.g., right ascension, azimuth etc.)

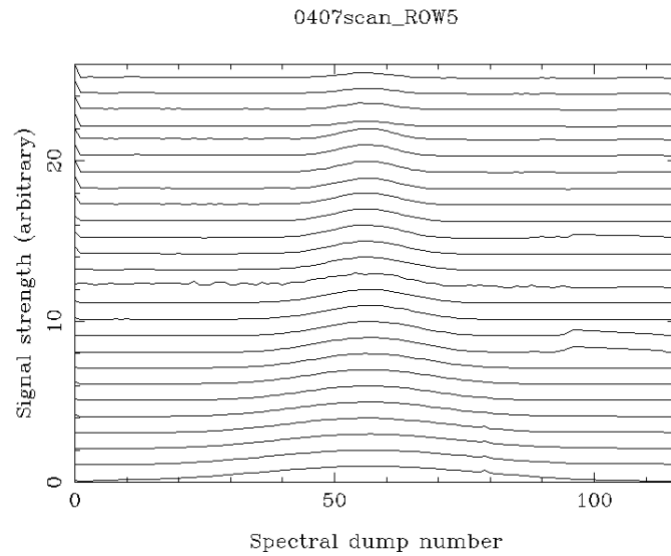
Clearly we need to carry out some RFI flagging as we are averaging in frequency across wide bands that contain RFI and we have at least one spectral dump affected by strong broadband interference. Note that this observation was unfortunately affected by: mobile handsets, aircraft DME, satellites and WiFi/Bluetooth!

We can do this with `sdhdf_flag` (see later tutorial, or the documentation, on the use of this package)

```
$ sdhdf_flag -f uwl_200720_024617.hdf
```

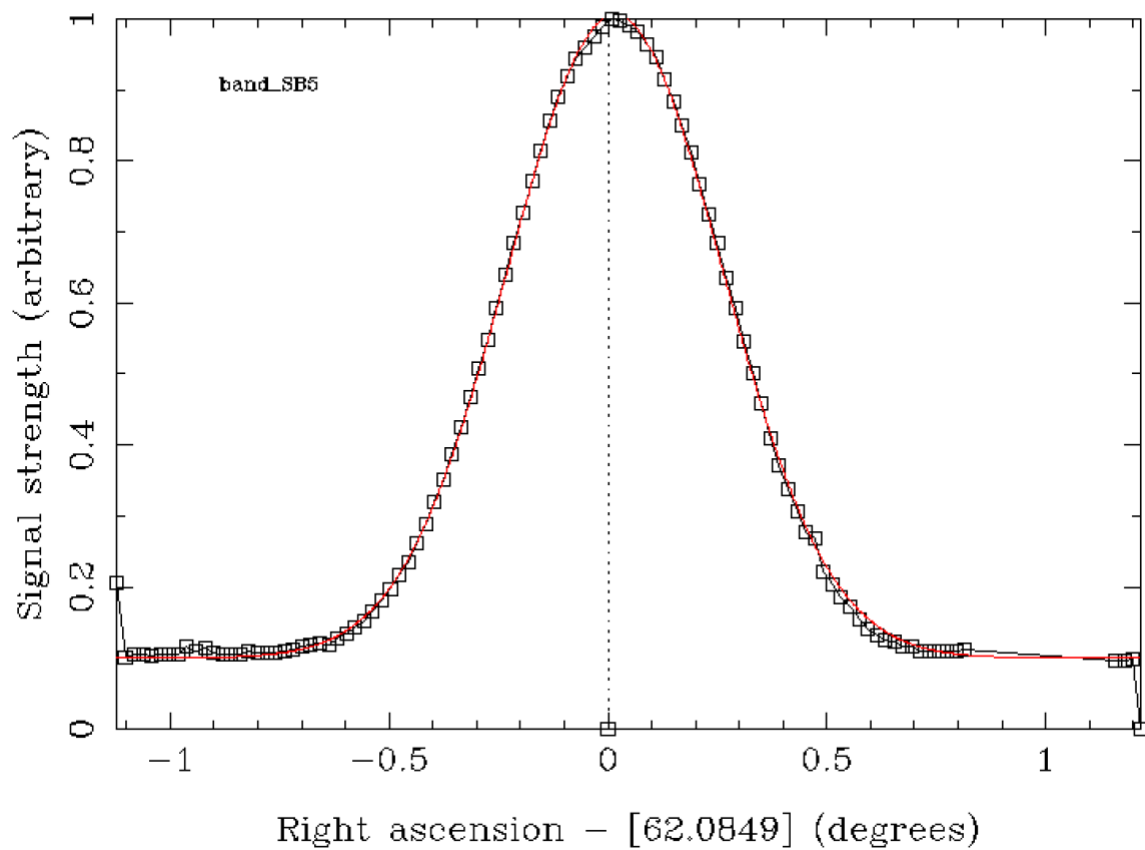
After flagging we can plot the scan again (note that we’re now using the `-src` option to specify the continuum source name):

```
$ sdhdf_plotScan -src 0407-658 -f uwl_200720_024617.hdf.flag
```



We can use the '+' button to select a specific sub-band and then press 's' to select that sub-band:

0407-658



Note that we've used 'x' to select a right ascension offset for the plot and 'f' to fit a Gaussian. The parameters for the fitted Gaussian (for each sub-band) are listed on the terminal.

Other Examples

Use case 1: On-off observations of a maser source (G323.459)

We first wish to extract (from the ON and OFF source pointings) the spectra covering the OH lines. We first copy all the relevant observations into the local directory. Then we make the files significantly smaller in data volume by using `sdhdf_extractBand` to form two zoom bands:

```
$ sdhdf_extractBand -zoom 1664 1668 -zoom 1718 1721 -e extractOH *.hdf
```

If we now inspect the content of the output files we get results similar to:

```
$ sdhdf_describe -band uwl_201027_045604.hdf.extractOH
```

File	Beam	UTC	SDHDF	Sched_ID	Source	Tel	Observer	RCVR	Bands	POL_TYPE	CAL
M_Time RA_s	DEC_s		PID								
uwl_201027_045604.hdf.extractOH		0	2020-10-27-04:56:04	1.9.3	P1073 18295	Off_320plus4	Parkes	gre469	UWL	2	
AABBCRCI	ON	237.9 14:53:28.00 -54:46:35.90									
#	BandID	F0	F1	NCHAN	TDUMP	NPOL	NDUMP	TOBS			
[Band] 000	band_zoom000	1664.00	1668.00	8192	0.983	4	242	237.896	uwl_201027_045604.hdf.extractOH		
[Band] 001	band_zoom001	1718.00	1721.00	6145	0.983	4	242	237.896	uwl_201027_045604.hdf.extractOH		

This shows that the output file now contains two sub-bands (labelled as `band_zoom000` and `band_zoom001`).

We now wish to flux calibrate the on- and off- source positions. As we have the calibrator running and 4-polarisation states we can directly from Stokes I:

```
$ sdhdf_modify -S -scal auto -T -e calibrated *.extractOH
```

We now wish to form the ON-OFF spectra (noting that we're still making use of topocentric frequencies at the moment) and so any RFI should be at the same frequency in both the ON and the OFF spectra.

```
$ sdhdf_identify -o match.dat -on G323p459 -off Off_320plus4 -pair *.extractOH.calibrated
$ sdhdf_onoff -batch match.dat -subtract
```

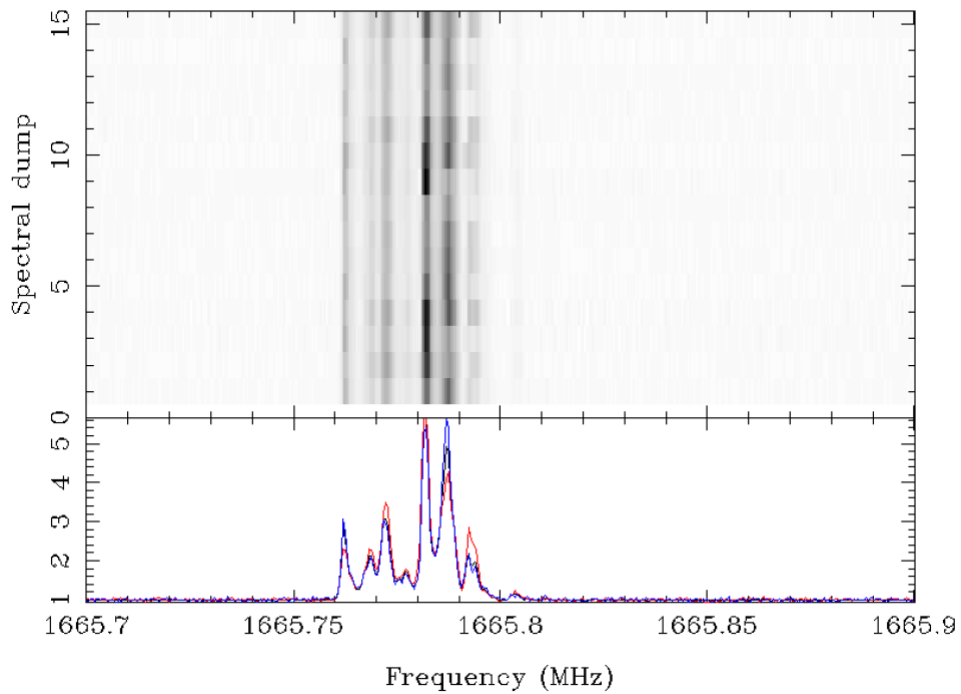
We can then convert the topocentric frequency axis to a local standard of rest frequency axis:

```
$ sdhdf_modify -lsr -regrid -e regrid *.onoff
```

Make a waterfall plot to ensure the frequency axis is correct

```
$ sdhdf_waterfall *.regrid -cleanRange 1665 1665.4 -frange 1665.7 1665.85
```

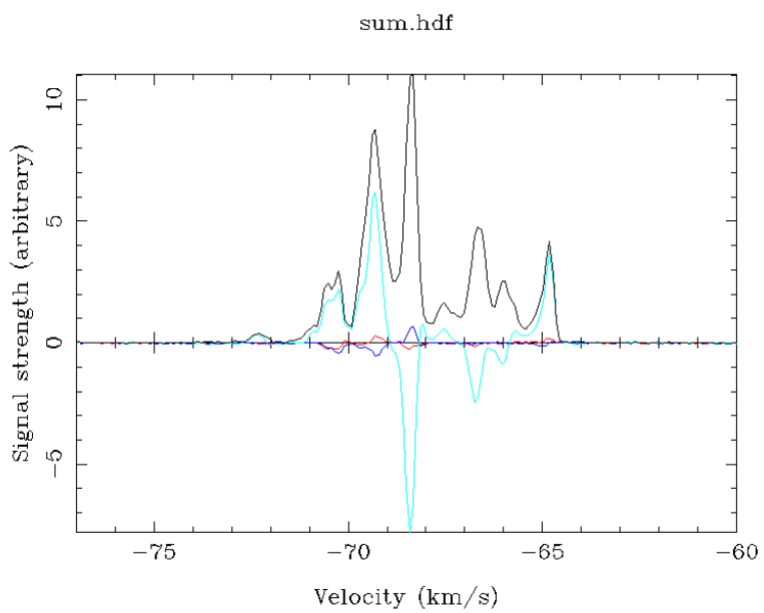
This produces a plot similar to the following:



We can average together all the spectra using:

```
$ sdhdf_sum -o sum.hdf -divN *.regrid  
$ sdhdf_plotMultiSpec -4pol -fref 1665.4019 -f0 -77 -f1 -60 -b1 -80 -75 sum.hdf
```

This gives (black = Stokes I, cyan = Stokes V, red and blue = Q and U):



Need to get the attributes correct

Use case 2: Scan across bright, continuum source

Have scan across e.g., Hydra A. Demonstrate that we can remove RFI, measure flux density (in Jy) versus offset position. Fit a beam shape to the resulting scan.

Should be able to do most of this now - but need new data files

uwl_191127_195059.hdf: P737 HydraA_DCSCAN

Use case 3: HI observation

Calibrate an HI spectrum in K and LSR velocity. Demonstrate that we can remove the baseline and compare with GASS. Have recent observation where the ripple will be lower.

Should be able to do now. 2020-01-04-14:15:15.hdf (P737).

Use case 4: Identifying spectral features

Use observation of Sgr region to identify multiple spectral lines. Calibrate in K.

Need suitable observation file.

Use case 5: Mapping

Goal: To create a map of the Cen A region

Data: uwl_190604_090701.hdf to uwl_190608_102736.hdf from P1017 (331 data files)

1. We first average all the spectral dumps within each file (COULD ALSO REMOVE CROSS-TERMS HERE):
> sdhdf_modify -T -e T uwl*.hdf
2. Let's check what files we have:
> sdhdf_describe *.T

The output is similar to:

```
-----
File                               UTC                               SDHDF PID   Source      Tel
Observer  RCVR    Bands POL_TYPE    CAL    Time
-----
uwl_190604_090701.hdf.T            2019-06-04T09:07:01Z            1.8   P1017   3c273_cal_R
Parkes    CA      UWL      26      AABBCRCI  ON      114.00
uwl_190604_090821.hdf.T            2019-06-04T09:08:21Z            1.8   P1017   3c273_cal_R
Parkes    CA      UWL      26      AABBCRCI  ON      95.00
uwl_190604_090945.hdf.T            2019-06-04T09:09:45Z            1.8   P1017   3c273_cal_R
Parkes    CA      UWL      26      AABBCRCI  ON      114.00
```

```

uwl_190604_091118.hdf.T      2019-06-04T09:11:18Z    1.8    P1017    3c273_cal12_R
Parkes      CA      UWL      26      AABBCRCI    ON      133.00
uwl_190604_091921.hdf.T      2019-06-04T09:19:21Z    1.8    P1017    cenA_ral_100
Parkes      CA      UWL      26      AABBCRCI    ON      437.00
uwl_190604_092328.hdf.T      2019-06-04T09:23:28Z    1.8    P1017    cenA_ral_101
Parkes      CA      UWL      26      AABBCRCI    ON      437.00
uwl_190604_092736.hdf.T      2019-06-04T09:27:36Z    1.8    P1017    cenA_ral_102
Parkes      CA      UWL      26      AABBCRCI    ON      437.00
...

```

3. To obtain an “average” spectrum for RFI identification we sum all the cenA observations together:

```
> sdhdf_sum -o cenA.sum `sdhdf_identify -src cenA -noinfo uwl*.hdf`
```

(this output file can be viewed using `sdhdf_plotSpectrum`).

4. Now let's flag the RFI:

```
> sdhdf_flag -f cenA.sum
```

Press 'b' to remove 5% of band edges

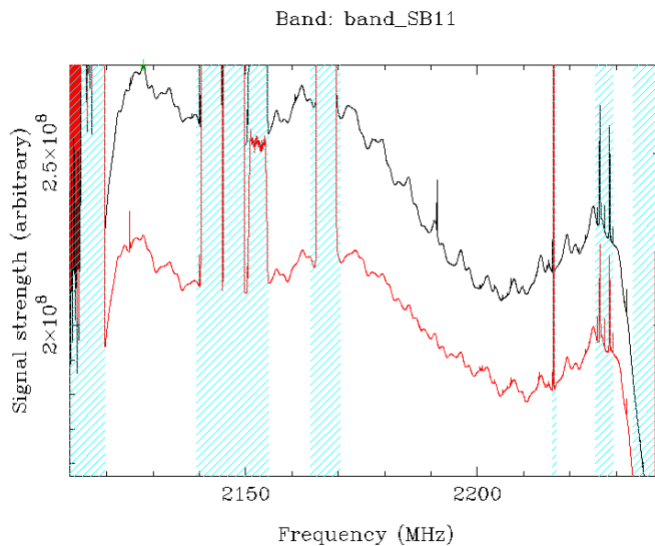
'z' to zoom-in ('u' to unzoom)

'Z' to flag a region.

'+' to move to the next sub-band

'-' to move to the previous sub-band

After flagging, the plots should be similar to:



Press 's' to save the flagged data file as `cenA.sum.flag`.

5. We wish to apply this flagging information to all our observations (note we need the scans here and so we use the original .hdf files):

```
> sdhdf_autoFlag -from cenA.sum.flag uwl*.hdf
```

This will produce multiple files with file extensions .autoflag

6. Should now set scalings based on closest primary calibrator (or using the cal etc.)

The calibrator observations of 3C273 are:

```
uwl_190604_090945.hdf
uwl_190605_090916.hdf
uwl_190606_092526.hdf
uwl_190607_091754.hdf
uwl_190608_091254.hdf
```

Let's average over all the spectral dumps in these files:

```
> sdhdf_modify -T -e T uwl_190604_090945.hdf uwl_190605_090916.hdf
uwl_190606_092526.hdf uwl_190607_091754.hdf uwl_190608_091254.hdf
```

Now we scale the flagged spectra into units of the primary calibrator

```
> sdhdf_scale -from uwl_190604_090945.hdf.T uwl_190604_*.hdf.autoflag
> sdhdf_scale -from uwl_190605_090916.hdf.T uwl_190605_*.hdf.autoflag
> sdhdf_scale -from uwl_190606_092526.hdf.T uwl_190606_*.hdf.autoflag
> sdhdf_scale -from uwl_190607_091754.hdf.T uwl_190607_*.hdf.autoflag
> sdhdf_scale -from uwl_190608_091254.hdf.T uwl_190608_*.hdf.autoflag
(should do 190609 and 190610 as well - no cals?)
```

7. Now let's make the map:

```
> sdhdf_map -sb 5 uwl*.hdf.autoflag.scale
(/u/hob044/software/new_c/sdhdfProc/sdhdf_map -sb 5
uwl_190604_*.hdf.autoflag.scale)
```

8. (/u/hob044/software/new_c/sdhdfProc/sdhdf_map -loadPlot mapGrid.dat)

Use case 6: Rise to set observations of PKS 1934

Manipulating files

It is possible to load in one, or more, input data files, carry out some processing steps and then output a new set of SDHDF files. The input files are listed on the command line and the output files have the same names, but with a user-defined file extension. For instance, the input file may be `uwl_191208_055418.hdf`, but after summing all the spectral dumps together the output file name may be `uwl_191208_055418.hdf.T`.

- `sdhdf_extract`: Extracts segments from a given file to produce a smaller output file. This routine can extract specific sub-bands or zoom-bands.
- `sdhdf_modify`: Modifies a data file by, e.g., averaging spectral dumps, forming Stokes parameters, or converting topocentric frequencies to barycentric or LSR frames.
- `sdhdf_sum`: Produces a single output file in which the spectra are the sum of all the spectra in the input files.

sdhdf_extractBand

Enables the copying of specific parts of an SDHDF file to a new SDHDF file. For instance, this code can be used to extract one or more sub-bands from a given file and output to a new file. In this example:

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_extractBand -e extract -b band_SB05 -b band_SB12  
input*.hdf
```

Subbands with labels `band_SB05` and `band_SB12` are obtained from files containing `input*.hdf` in their name and written to the filename with extension `.extract` (defined by the `-e` option). Note that if the band is simply identified with an integer then the band number (starting from 0) in the file is used instead.

It is also possible to extract one or more zoom-bands from a data file (including over sub-band boundaries):

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_extractBand -e extract -zoom 1420 1424 input*.hdf
```

Will extract a single zoom band covering frequencies from 1420 to 1424 MHz (defined by the `-zoom <f0> <f1>` option).

sdhdf_modify

Carries out simple modification/processing routines on an SDHDF file and produces a new one. Options:

`-e` output file extension

-T	sum all spectral dumps
-bary	convert frequencies to barycentric frequencies
-lsr	convert frequencies to the local standard of rest
-fav <integer value>	average 'value' spectral channels together
-F	average to a single frequency channel
-p1	form total intensity: (Pol 1 + Pol 2)/2. (note dividing output by 2)
-p2	keep on Pol 1 and Pol 2 (not cross terms)
-p3	form total intensity: (Pol 1 + Pol 2) (note not dividing output by 2)
-S	transform to Stokes parameters (note, assuming already calibrated)

For example we wish to sum the spectral dumps in uwl_191208_055418.hdf:

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_modify -e T -T uwl_191208_055418.hdf
```

The output is a new sdhdf file called uwl_191208_055418.hdf.T which can then be processed as normal (e.g., using plotting routines).

sdhdf_sum

Software to sum together (and produce an average) of the spectra in multiple files.

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_sum -o outFile.hdf in*.hdf
```

Note that currently this only works with:

1. 1 spectral dump per file
2. All files with the same number of frequency channels and sub-bands
3. 4 pol data

Flags

The sdhdf weight information provides a relative weight of the amount of data combined into a specific spectral channel and spectral dump. If the table is not set then the weight values will be set to the product of the integration time (in seconds) corresponding to the spectral dump and the channel bandwidth (in Hz). When frequency channels or time dumps are averaged then the weights sum. If a spectral dump or frequency channel is flagged then the corresponding weight is set to zero.

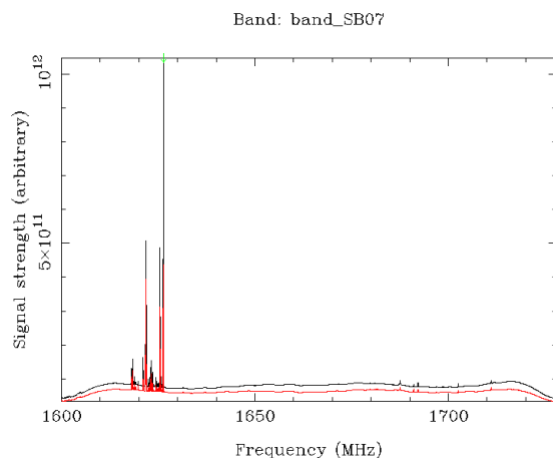
~~A flag can be set for each channel in a given observing band. A flag = 1 implies that the spectral channel should be discarded. A flag = 0 implies that the spectral channel should be used when plotting and processing. Note that the same flag is used for all spectral dumps and so cannot be used to de-weight bad spectral dumps.~~

sdhdf_flag

We wish to flag uwl_190608_102736.hdf.

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_flag -f uwl_190608_102736.hdf
```

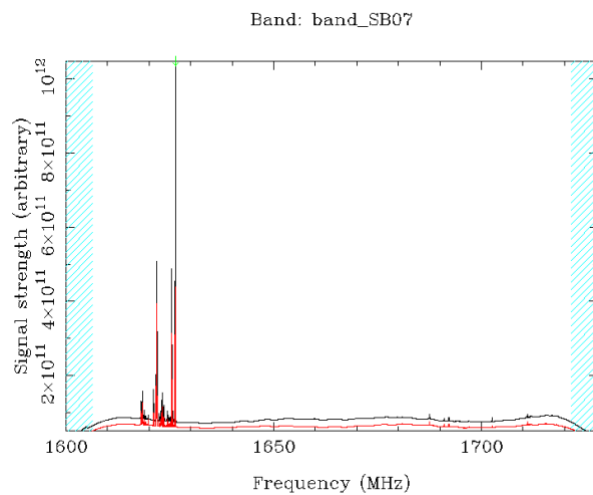
The time-averaged spectra for a given sub-band are plotted:



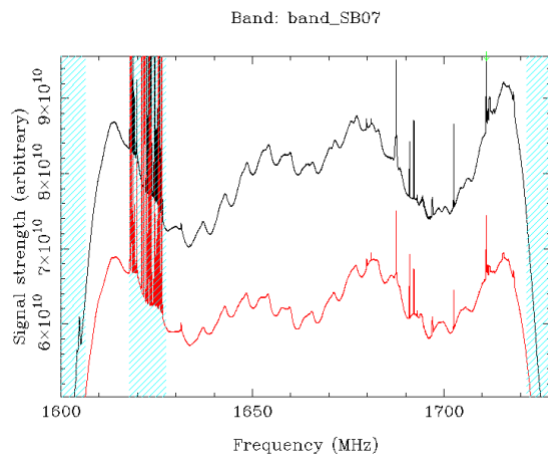
Key presses for general use are:

- '+' move to next sub-band
- '-' move to previous sub-band
- 'z' use the mouse to select a region to zoom-in on.
- 'u' unzoom
- 'q' quit

The user can also press 'b' which will automatically zap 5% of the Parkes UWL sub-band edges:



Flagged regions are shown using the light-blue, hashed regions. Other regions to flag can be selected manually by pressing 'Z' at the left-hand edge of the region, moving the mouse to the right-hand edge and then clicking the left-mouse button.



The plot automatically re-scales on the regions that have not been flagged.

Press 's' to save a new output file (with extension .flag) that will contain the flagging information.

(Note to aid the user for manual flagging the strongest RFI peak in a given plot is identified with a green, downward-pointing arrow at the top of the plot.)

sdhdf_autoFlag

This program enables the user to flag a set of data files automatically. The only routine implemented so far copies the flag information from a specific file (specified using the -from command line argument) to all the other files listed on the command line.

```
% sdhdf_autoFlag -from cenA.sum.flag uwl*.hdf
```

NOT IMPLEMENTED YET

1. Automatically zap N% of band edges
2. Automatically zap known u
3. Read in lots of files and "OR" the flags.

Converting to and from SDHDF files

sdhdf_convertFrom

This routine converts spectral line data formats into the SDHDF format from a different format:

- Text files
- SDFITS (NOT IMPLEMENTED YET)

The text file format is simply two columns (currently only 1 pol data implemented so far):
<freq (MHz)> <signal>

Let's assume that the input text file is called psd.dat:

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_convertTo psd.dat
```

This produces an output file psd.dat.hdf which can be processed as normal using the tools described in this document.

sdhdf_convertTo

The routine will convert SDHDF files into other data formats. It currently converts to SDFITS format (as defined by https://casa.nrao.edu/aips2_docs/notes/236/node14.html).

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_convertTo uwl_190608_102736.hdf
```

The input file names are given as command line arguments. The output file names are the same as the input with extension .sdfits. Other command line arguments are:

- -e set the output file extension (default .sdfits)
- -1scanPerFile define all the spectral dumps in a given file as a single scan
- -srcChange <x> <y> change the source name "x" into "y" when writing the SDFITS file
- -band <x> only output band <x> to the SDFITS file
- -o <filename> output all data into a single output file: filename
- -h help information

Note that this conversion code currently does not convert calibration information.

Spectral line/continuum processing tools

sdhdf_onoff

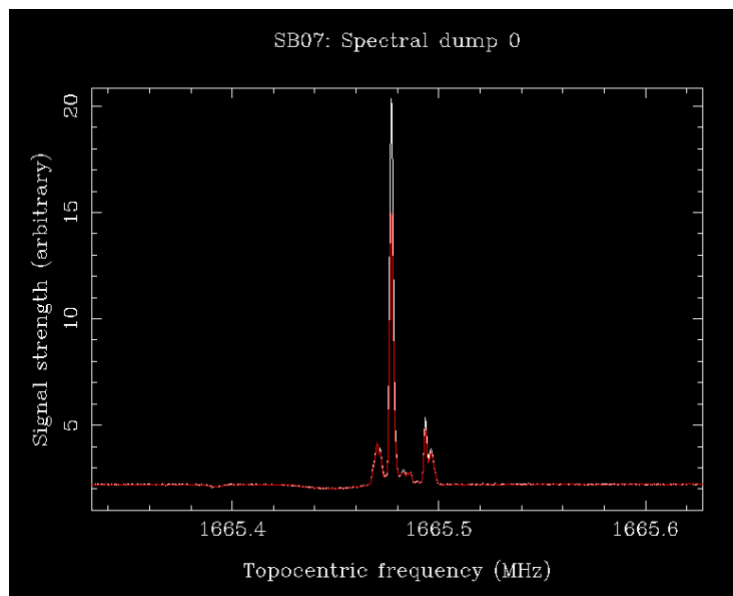
Takes a source observation (ON) and a reference observation (OFF) and produces a new file that contains (ON-OFF)/OFF.

Usage:

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_onOff -on onFile.hdf.T -off offFile.hdf.T -o diff.hdf
```

This produces diff.hdf which contains the difference (ON-OFF)/OFF for each sub-band. The output can then be plotted using sdhdf_plotSpectrum:

```
/u/hob044/software/new_c/sdhdfProc/plotSpectrum -f diff.hdf -sb 0
```



Other sdhdf_onOff command line arguments include:

- -scIA [X] A scaling factor to multiple the output spectrum for Pol A: X(ON-OFF)/OFF
- -scIB [X] A scaling factor to multiple the output spectrum for Pol B: X(ON-OFF)/OFF

It is common to process multiple files simultaneously. To do this, produce a text file with three columns giving the ON, OFF and output filenames, e.g.:

```
uwl_191213_130258.hdf.T uwl_191213_123813.hdf.T s32_01.q
uwl_191214_093035.hdf.T uwl_191214_091041.hdf.T s32_02.q
uwl_191214_095559.hdf.T uwl_191214_101555.hdf.T s32_03.q
uwl_191214_124606.hdf.T uwl_191214_122628.hdf.T s32_04.q
```

```
uwl_191215_082415.hdf.T uwl_191215_085605.hdf.T s32_05.q  
uwl_191216_075017.hdf.T uwl_191216_090232.hdf.T s32_06.q  
uwl_191216_083041.hdf.T uwl_191216_090232.hdf.T s32_07.q
```

Then we use a command line similar to the following (where onOff_list_s32 is the filename of the file given above):

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_onoff -batch onOff_list_s32
```

Current issues include:

- Requires only 1 spectral dump in both on and off files
- Does not use flags

sdhdf_map

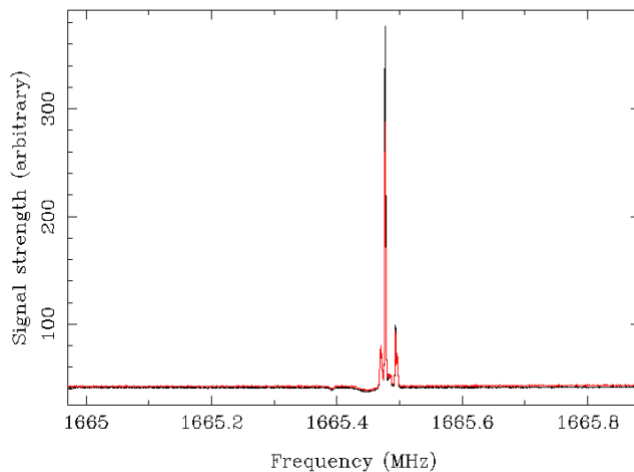
sdhdf_map is used to make a sky map from a set of scans.

```
/u/hob044/software/new_c/sdhdfProc/sdhdf_map -o map.hdf <filenames>
```

sdhdf_baseline

This code is used to determine (and remove) baselines from spectra.

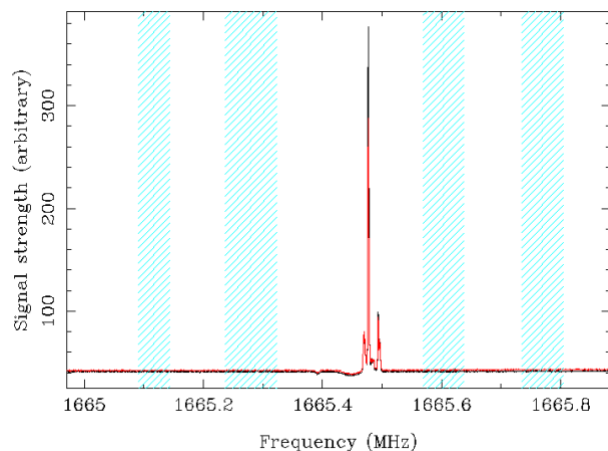
```
/u/hob044/software/new_c/sdhdfProc/sdhdf_baseline -f <filename.hdf>
```



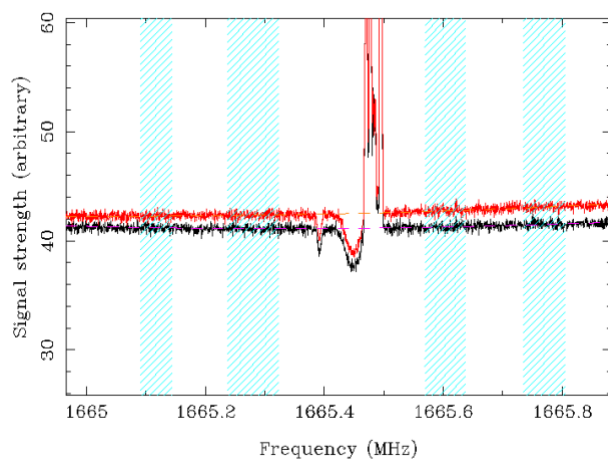
Standard key-strokes can be used to inspect the plot:

- ‘z’ - zoom in on a region
- ‘u’ - unzoom
- ‘q’ - quit

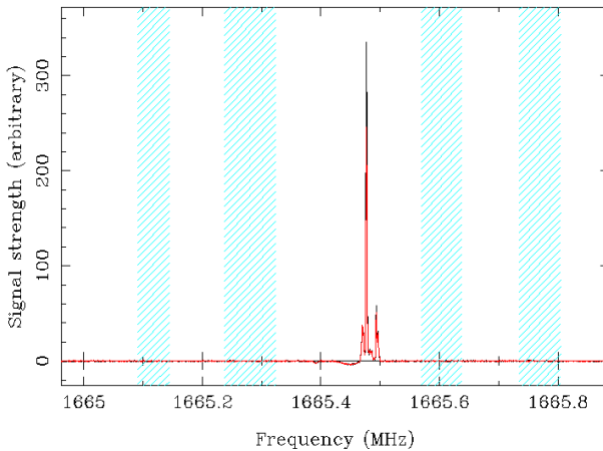
Pressing 'b' allows one or more baseline regions to be set. Press 'b' when the cursor corresponds to the lowest frequency in the baseline region and then click the left mouse button at the highest frequency. Press 'b' again to repeat for a different region. The selected regions will be highlighted:



Press 'n' to select the number of polynomial coefficients to include in the polynomial fit (1 = just mean, 2 = quadratic etc.). Then press 'f' to fit the polynomial coefficients:



The fitted function (for Pol A and Pol B) are shown as dashed lines. Press 'r' to remove the fitted function from the data.



sdhdf_pointing

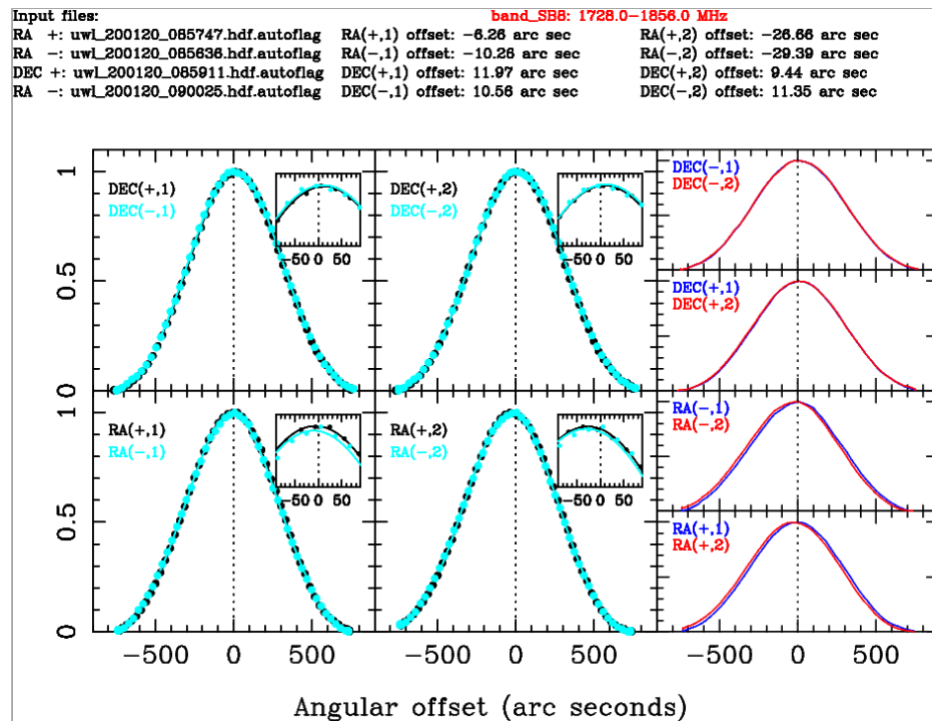
sdhdf_map is used to make pointing solutions using multiple scans across known sources.

Note that much of this code is currently hardcoded to the Parkes telescope. The code inputs a scan across a known source in right ascension in the forward direction, then again in the backward direction and then the same in declination.

Typical usage is:

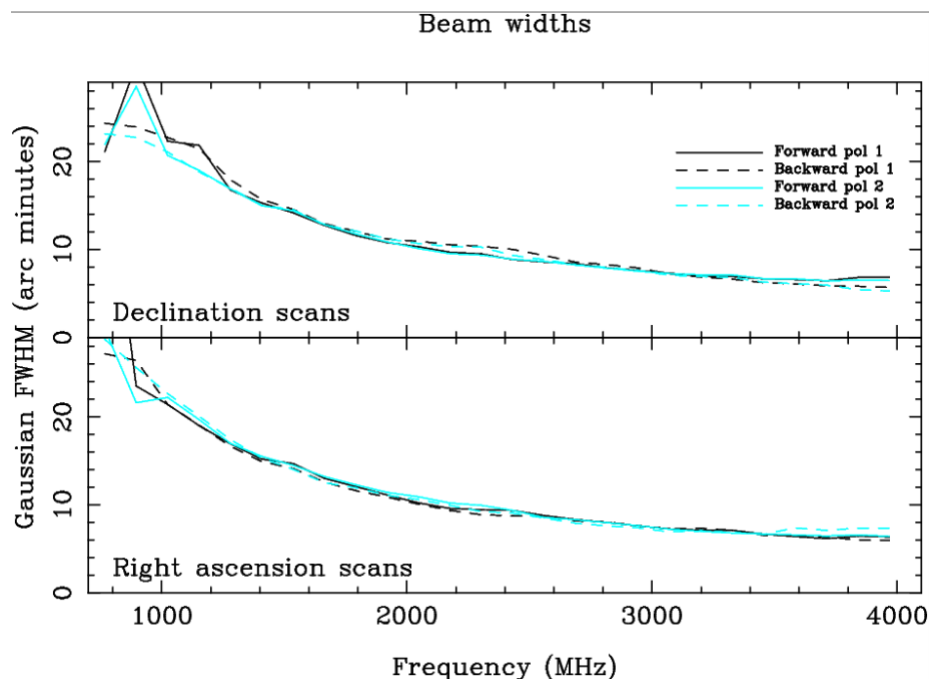
```
/u/hob044/software/new_c/sdhdfProc/sdhdf_pointing -raf uwl_200120_085747.hdf.autoflag -rab  
uwl_200120_085636.hdf.autoflag -decf uwl_200120_085911.hdf.autoflag -decb  
uwl_200120_090025.hdf.autoflag
```

Note that the data files have already been flagged for RFI (either manually or automatically). In each sub-band the code determines the signal strength versus time. The output is in sdhdf_pointing.ps. For the Parkes UWL receiver the first 26 pages in the output postscript file correspond to the 26 subbands:



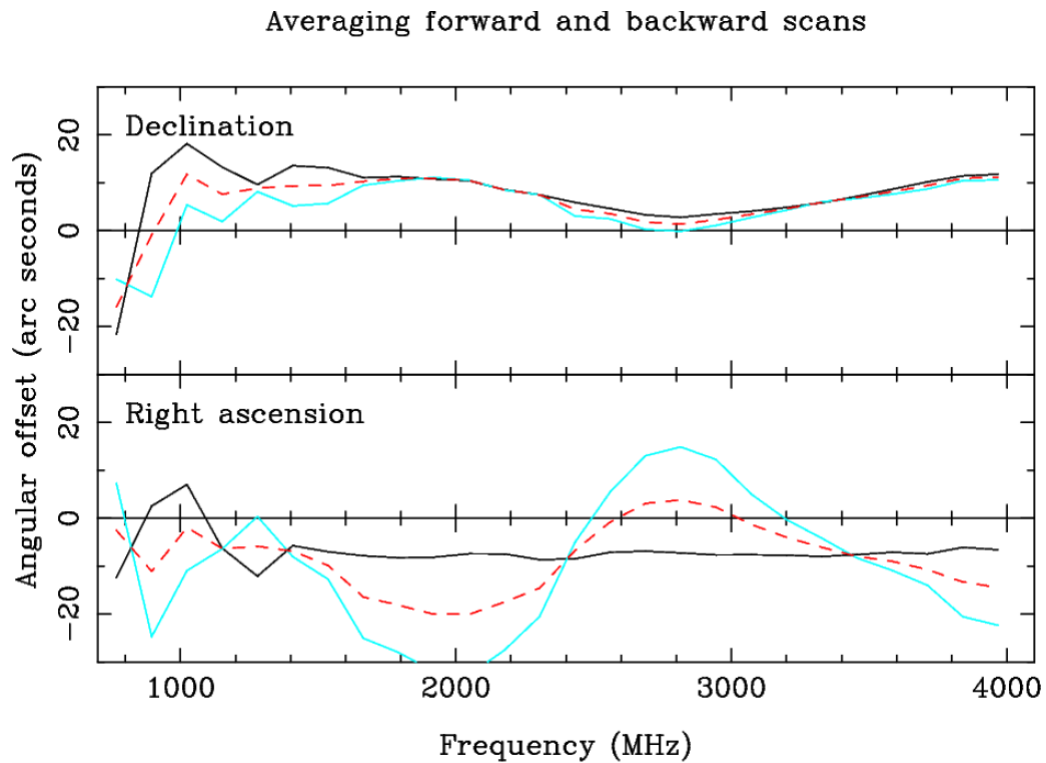
The panels show the normalised signal strength for two polarisations and for the forward and backward directions. The labels indicate RA/DEC(FWD/BACK, Pol1/2). A Gaussian function is fitted to the central regions and the offset from the known source position is listed in the values at the top of the page.

The beam widths are presented on page 27:



And these widths used to derive the antenna gain (in K/Jy) and the antenna efficiency:

The final page shows the pointing offsets:



A .pop file is written out for each sub-band to disk. These files contain the following columns:

dRA(s) dDEC(") HA_hh HA_mm DEC_hh DEC_mm Source FileName RCVR Para AXD ELD

Using the calibrator

The Parkes UWL calibrator can be “ON”, “OFF” or “SWITCHING”. If the calibrator is switching then the Medusa GPU cluster produces (1) a CAL-ON spectrum, (2) a CAL-OFF spectrum and (3) the astronomy spectrum. The CAL-ON and CAL-OFF spectra usually have 128 channels/sub-band (1MHz channels) and a spectral dump time of 5 seconds. The scaling factor between the values (V) in the spectra (measured in counts) for the CAL spectra and the astronomy spectra is:

$$V_{astro} = V_{cal} \frac{N_{chan,astro}}{N_{cha,cal}} \frac{1}{2} \times \frac{t_{astro}}{t_{cal}}$$

where t_{astro} and t_{cal} are the dump times for the astronomy spectrum and calibrator spectra respectively. (Note, I think that the factor of 0.5 comes from the fact that only half the signals are in the ON or the OFF spectra compared with the astronomy spectrum).

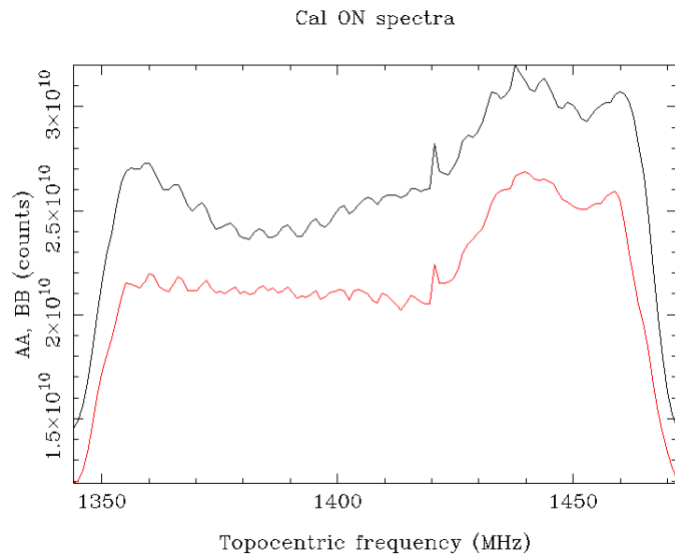
NEED TO HAVE THE METADATA CORRECT FOR SUBINT LENGTHS ETC. BEFORE APPLYING THE CAL AS NEED TO KNOW t_{astro}

sdhdf_cal

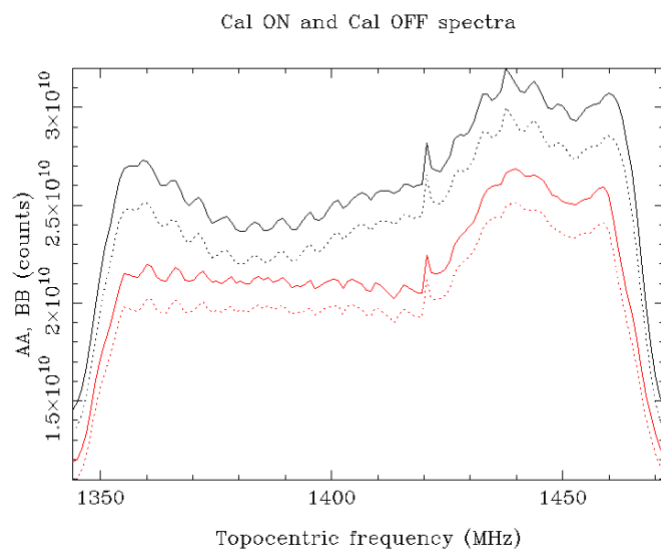
```
/u/hob044/software/new_c/sdhdfProc/sdhdf_cal -sb 1 -f uwl_191205_102837.hdf.T
```

This code produces various plots. Press the numbers 1 to 9 to select different plots. ‘z’ will zoom-in on a plot, ‘u’ will unzoom and ‘m’ will print out the coordinates of a given mouse click.

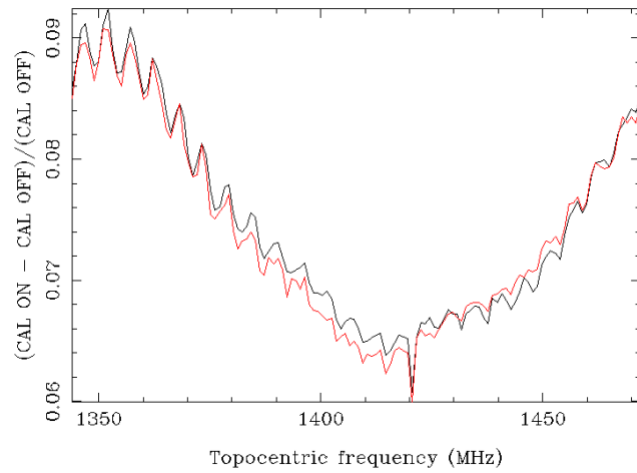
Plots 1 and 2 will show the CAL ON and CAL OFF spectra respectively:



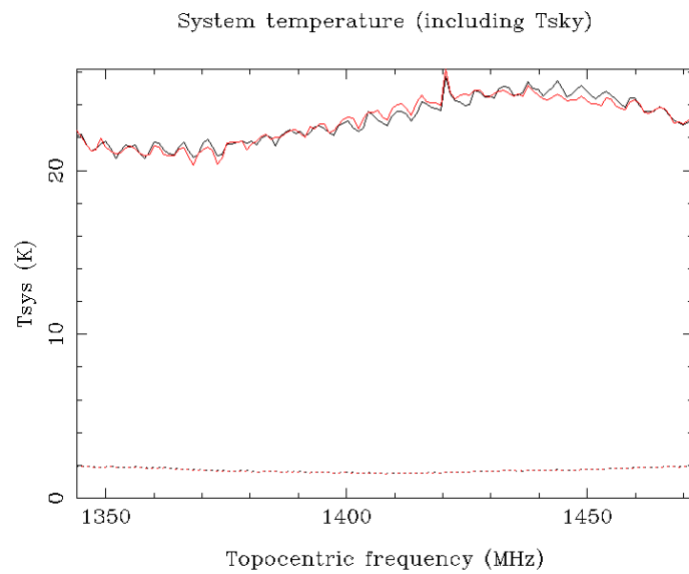
Plot '3' plots the CAL ON (solid) and the CAL OFF (dotted) spectra on the same picture:



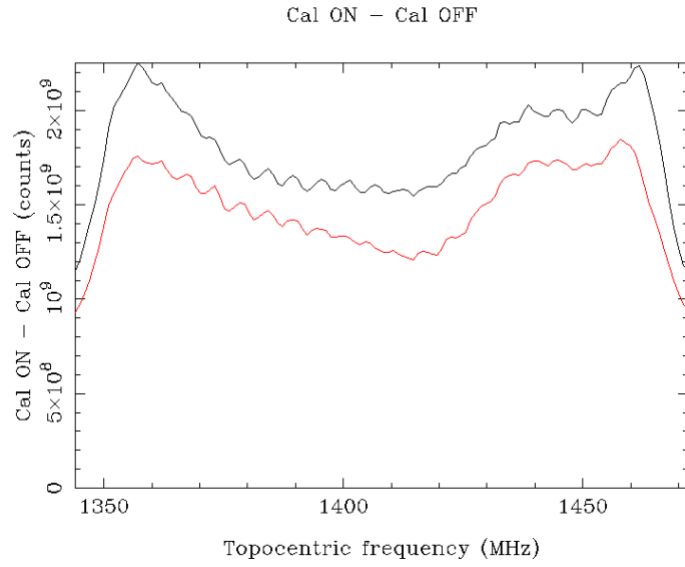
Plot '4' shows $(\text{CAL ON} - \text{CAL OFF}) / (\text{CAL OFF})$:



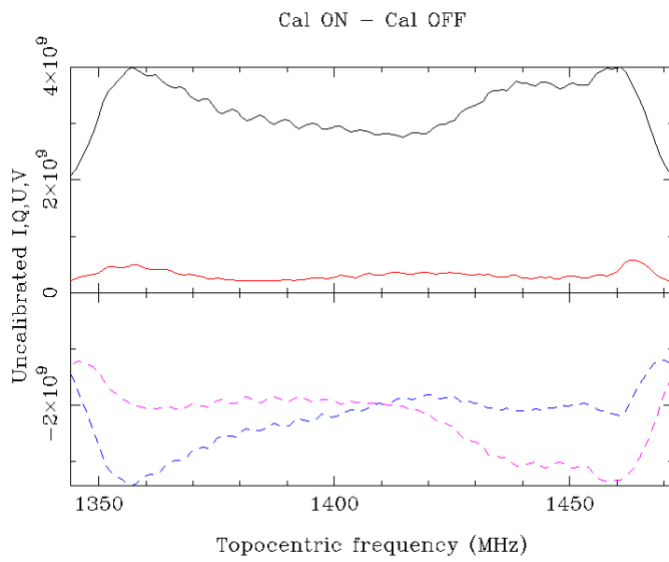
This is converted to T_{sys} , using knowledge of $T_{\text{cal}}(f)$ which is shown as the dotted line in the bottom of the figure, in plot '5':



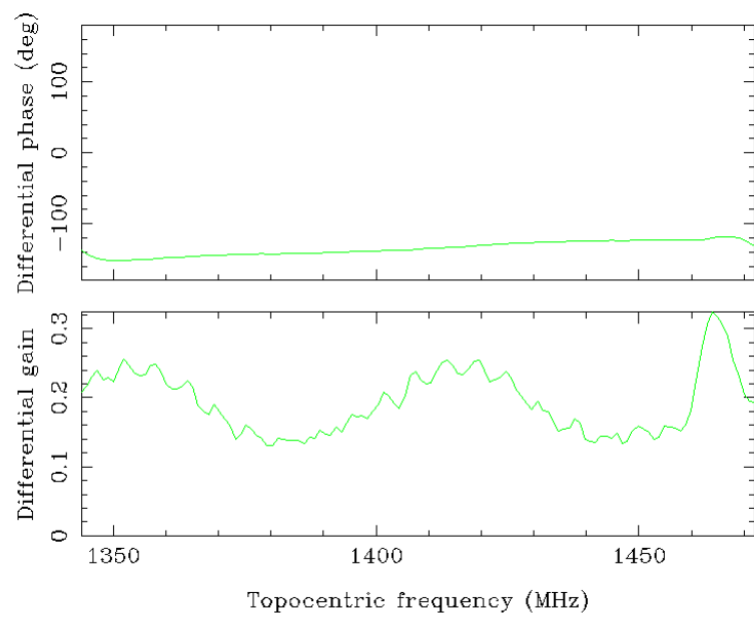
Plot '6' shows $(\text{CAL ON} - \text{CAL OFF})$ in units of counts:



The calibrator signal is shown in “uncalibrated Stokes parameters” in plot ‘7’:



The differential gain and phase as measured from the calibrator is plotted in ‘8’:



Miscellaneous code

sdhdf_listLines

This code displays the rest frequencies of known lines in the UWL band:

```
sdhdf_listLines <f0 (MHz)> <f1 (MHz)>
```

This gives an output similar to:

1420.4058	HI: Neutral hydrogen
1538.1080	NH_2CHO
1538.6760	NH_2CHO
1539.2640	NH_2CHO
1539.5270	NH_2CHO
1539.8320	NH_2CHO
1540.9980	NH_2CHO
1570.8051	NH_2 ¹³ CHO
1584.2740	¹⁸ OH
1610.2469	CH_3OCHO
1610.9000	CH_3OCHO
1612.2310	OH: Hydroxyl radical
1624.5179	¹⁷ OH
1626.1610	¹⁷ OH
1637.5640	¹⁸ OH
1638.8051	HCOOH
1639.5031	¹⁸ OH
1665.4019	OH: Hydroxyl radical
1667.3590	OH: Hydroxyl radical
1692.7950	¹⁸ OH

sdhdf_identify

This code allows the user to identify SDHDF files in which a particular source was observed, or the observation covered a specific sky region:

```
% /u/hob044/software/new_c/sdhdfProc/sdhdf_identify -src cenA uwl*.hdf
```

will list the filenames containing the source “cenA”:

```
[SRC MATCH] uwl_190604_091921.hdf cenA_ra1_100
[SRC MATCH] uwl_190604_092328.hdf cenA_ra1_101
[SRC MATCH] uwl_190604_092736.hdf cenA_ra1_102
[SRC MATCH] uwl_190604_093144.hdf cenA_ra1_103
...
```

Use -noinfo if you only wish to obtain the filenames:

```
uwl_190604_091921.hdf
uwl_190604_092328.hdf
uwl_190604_092736.hdf
uwl_190604_093144.hdf
```

To identify all the files in which a pointing was within ‘D’ degrees of a specified RA/Dec coordinate (RA,DEC) then use

```
% /u/hob044/software/new_c/sdhdfProc/sdhdf_identify -coord <RA> <DEC> <D> uwl*.hdf
```

For instance:

```
% /u/hob044/software/new_c/sdhdfProc/sdhdf_identify -coord 198.068 -44.8243 0.1 uwl*.hdf
```

gives:

```
[DIST MATCH] uwl_190604_105832.hdf 318 0.0600269
[DIST MATCH] uwl_190604_105832.hdf 319 0.098256
[DIST MATCH] uwl_190604_110240.hdf 120 0.0882875
[DIST MATCH] uwl_190604_110240.hdf 121 0.0442946
```

The third column gives the spectral dump number corresponding to the angular offset (in degrees) given in the fourth column.